💻 Terminal Commands Guide (Windows & macOS)

A complete list of essential terminal commands from beginner to advanced for both Windows (CMD/PowerShell) and macOS/Linux.

🟢 Basic Navigation

Action	Windows (CMD/PowerShell)	macOS / Linux
List files	dir	ls
Change directory	cd folder_name	cd folder_name
Go back	cd ..	cd ..
Clear screen	cls	clear
Show path	cd	pwd

📁 File & Folder Management

Action	Windows	macOS / Linux
Create folder	mkdir folder_name	mkdir folder_name
Delete folder	rmdir folder_name	rm -r folder_name
Create file	echo text > file.txt	touch file.txt
Delete file	del file.txt	rm file.txt
Copy file	copy a.txt b.txt	cp a.txt b.txt
Move file	move a.txt folder	mv a.txt folder

📄 File Viewing

Action	Windows	macOS / Linux
View file	type file.txt	cat file.txt
View large file	more file.txt	less file.txt

🔍 Search & Find

Action	Windows	macOS / Linux
Search text	find "text" file.txt	grep "text" file.txt
Find files	dir /s filename	find . -name filename

🌐 Network Commands

Action	Windows	macOS / Linux
Check connection	ping google.com	ping google.com
Show IP	ipconfig	ifconfig / ip a

⚙️ System Commands

Action	Windows	macOS / Linux
Show processes	tasklist	ps
Kill process	taskkill /PID 1234	kill 1234

🔥 Advanced Commands

🧠 PowerShell (Windows Advanced)

```
Get-ChildItem        # List files (like ls)
Set-Location folder    # Change directory
Remove-Item file.txt   # Delete file
```

🧠 macOS/Linux Advanced

```
chmod +x script.sh     # Make file executable
sudo command          # Run as admin
history             # Show command history
```

🔁 Git + Terminal Workflow Example

```
cd project-folder
git init
git add .
git commit -m "first commit"
git branch -M main
git remote add origin <repo-url>
git push -u origin main
```

⚠️ Tips for Beginners

- Use `Tab` for auto-complete
- Use ↑ arrow for previous commands
- Be careful with `rm -rf` (deletes permanently ⚠️)
- Use `clear` or `cls` to keep terminal clean

🎯 Summary

- Windows → `dir`, `copy`, `del`
- macOS/Linux → `ls`, `cp`, `rm`
- Git works same on both platforms

👩‍💻 Real Developer Workflows

🚀 1. Starting a New Project

```
mkdir my-project
cd my-project
git init
touch index.html style.css
git add .
git commit -m "Initial project setup"
```

🔄 2. Daily Work Routine

```
git pull origin main      # Get latest changes
# make changes in code
git add .
git commit -m "Updated feature"
git push origin main
```

🌿 3. Feature Branch Workflow (Best Practice)

```
git checkout -b feature-login
# work on feature
git add .
git commit -m "Added login feature"
git push origin feature-login
```

Then:

- Create Pull Request on GitHub

- Merge into `main`

🔥 4. Fixing Mistakes

```
git restore file.txt        # Undo changes
git reset --soft HEAD~1     # Undo commit (keep code)
git reset --hard HEAD~1     # Delete commit (danger ⚠)
```

🧳 5. Switching Work (Stash)

```
git stash
git checkout another-branch
git stash pop
```

📊 Mini Practice Projects

💡 Project Ideas

- 📝 Notes App (HTML + CSS)
- 📋 To-Do List
- 🌐 Personal Portfolio
- 🎨 Landing Page

📁 Suggested Structure

```
project-name/
 |── index.html
 |── style.css
 |── script.js
 |── README.md
```

⚡ Productivity Tips

- Press `Tab` → auto-complete commands
- Use `↑` → repeat last command
- Use `history` → see all commands
- Keep commits small & meaningful

🧩 Common Mistakes (Avoid These)

❌ Pushing without pulling ❌ Using `--force` without understanding ❌ Working directly on `main` ❌ Not writing commit messages

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🏆 Pro Level Tips

- Use branches for every feature
- Write clean commit messages
- Keep repo organized
- Practice daily using Git

🎯 Final Goal

✔️ Understand Git basics ✔️ Work with branches ✔️ Collaborate using GitHub ✔️ Build real projects

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⭐ Bonus

If this helped you:

- Star ⭐ the repo
- Share with friends
- Keep practicing Git daily 🚀

Mastering terminal commands = faster development 🚀